A COVERING OF THE GENERIC FIBRE BY CLOSED SUBSETS SPREADS TO A NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6657-6666 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6668-6683, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Closures of closed subsets in the total space, fibres of a finite type morphism, and the spreading of finitely many conditions; the argument is retained.

Source reference. Original lemma 054W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6496–6501. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03HV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6683–6683. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 052A: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6683–6683. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ON: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6683–6683. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a finite type morphism of schemes. Assume Y irreducible with generic point η. Let $X_\eta = Z_{1,\eta} \cup \ldots \cup Z_{n,\eta}$ be a covering of the generic fibre by closed subsets of X_η. Let Z_i be the closure of $Z_{i,\eta}$ in X (see discussion above). Then there exists a nonempty open $V \subset Y$ such that $X_y = Z_{1,y} \cup \ldots \cup Z_{n,y}$ for all $y \in V$.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. If Y is Noetherian then $U = X \setminus (Z_1 \cup \ldots \cup Z_n)$ is of finite type over Y and we can directly apply Lemma 054W to get that $U_V = \emptyset$ for a nonempty open $V \subset Y$. In general we argue as follows. As the question is topological we may replace Y by its reduction. Thus Y is integral, see Properties, Lemma 01ON. After shrinking Y we may assume that $X \to Y$ is flat, see Morphisms, Proposition 052A. In this case every point x in X_y is a specialization of a point $x' \in X_\eta$ by Morphisms, Lemma 03HV. As the Z_i are closed in X and cover the generic fibre this implies that $X_y = \bigcup Z_{i,y}$ for $y \in Y$ as desired. $\square$